Adaptive Per-Distance Sampling for Efficient Multi-Scale Network Centrality Analysis

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Abstract

Computing network centrality at scale is expensive. Sampling can help, but how much is enough? We present a practical model: maintain an effective sample size of $n_{\text{eff}} = \max(k \cdot \sqrt{r}, n_{\min})$, where r is network reach, to achieve ranking accuracy $\rho \geq 0.95$. The proportional term ($k = 14.21$) handles large networks; the floor ($n_{\min} = 350$) ensures reliability at low reach. The model validates on a 294,000-node Greater London street network, achieving $\rho > 0.98$ at distances from 5–20km with speedups of 3.7–14.6×. The implementation is available in the open-source `cityseer` Python library.

Keywords: network centrality, sampling, urban morphology, betweenness, harmonic closeness, computational efficiency

1 Introduction

Network centrality measures are fundamental tools for understanding urban street networks, revealing patterns of accessibility, connectivity, and flow that shape how cities function. Harmonic closeness centrality quantifies how effectively a location can reach all others, while betweenness centrality captures the extent to which a location lies on shortest paths between other pairs of nodes. These measures have proven invaluable for urban planning, retail location analysis, and understanding pedestrian movement patterns.

However, computing network centrality at scale presents a significant computational challenge. Standard algorithms have time complexity $O(N \cdot M + N^2 \log N)$ for weighted graphs, where N is the number of nodes and M is the number of edges. For large urban networks—which routinely contain hundreds of thousands of nodes—these computations become prohibitively expensive, particularly when analysis must be repeated across multiple distance thresholds or network configurations.

Sampling-based approximation offers a natural solution: rather than computing exact centrality from all source nodes, we compute from a random subset and scale the results accordingly. This approach can dramatically reduce computation time while preserving the essential ranking information that practitioners need. The critical question is: *how much sampling is enough?*

This paper presents a principled answer to that question. We introduce a simple, empirically-validated model for determining the appropriate sampling probability:

$$n_{\text{eff}} = \max\left(14.21 \cdot \sqrt{r}, 350\right), \quad p = \min\left(1, \frac{n_{\text{eff}}}{r}\right) \tag{1}$$

where r is the network reach (mean number of nodes within the analysis distance), n_{eff} is the effective sample size, p is the sampling probability, $k = 14.21$ is a fitted proportional constant, and 350 is a minimum floor ensuring reliability at low reach.

The model has two regimes. When reach exceeds 607, the proportional term dominates: the required effective sample size is $14.21 \cdot \sqrt{r}$, yielding sampling probability $p = 14.21/\sqrt{r}$ that decreases as networks grow larger. When reach is below 607, the floor dominates: we require an effective sample size of at least 350 to ensure statistical reliability.

Our contributions are:

1. **A practical sampling model**: We derive and fit a two-parameter model that guarantees ranking accuracy $\rho \geq 0.95$ across diverse network topologies and analysis distances. The model uses proportional scaling ($k \cdot \sqrt{r}$) with a minimum floor (350) to handle low-reach edge cases.

2. **Real-world validation**: We validate the model on a 294,000-node Greater London street network at distances from 5km to 20km, achieving speedups of 3.7–14.6× while maintaining $\rho > 0.98$.

3. **Open-source implementation**: The model is implemented in the `cityseer` Python library, enabling automatic per-distance sampling calibration with a single parameter.

The remainder of this paper is organised as follows. Section 2 reviews related work on network centrality and sampling approximation. Section 3 describes our experimental methodology. Sections 4 and 5 present the empirical models and adaptive sampling algorithm. Section 6 validates the approach on real-world networks. Section 7 provides practical guidance for practitioners. Section 8 discusses limitations, and Section 9 concludes.

2 Background and Related Work

2.1 Network Centrality in Urban Analysis

Network centrality measures have become essential tools for urban morphology and spatial analysis. Originally developed in social network analysis (**??**), these measures were adapted for urban street networks by **?** and subsequently formalised in the space syntax tradition.

Closeness centrality measures how efficiently a node can reach all others. The harmonic variant, which we use throughout this paper, is defined as:

$$C_H(v) = \sum_{u \neq v} \frac{1}{d(v,u)} \tag{2}$$

where $d(v,u)$ is the shortest path distance from v to u. The harmonic formulation handles disconnected graphs naturally and is preferred for distance-weighted urban analysis (**?**).

Betweenness centrality measures the extent to which a node lies on shortest paths between other pairs:

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}} \tag{3}$$

where σ_{st} is the number of shortest paths from s to t, and $\sigma_{st}(v)$ is the number passing through v. High betweenness indicates locations that control flow through the network.

For urban applications, these measures are typically computed with a distance threshold $d_{\max}$, considering only nodes within a specified radius. This reflects the reality that pedestrian accessibility is local rather than global. Common thresholds range from 400m (5-minute walk) to 20km (regional connectivity).

2.2 Computational Complexity

The computational cost of centrality measures scales poorly with network size. Brandes' algorithm (**?**) computes both betweenness and harmonic closeness in $O(NM + N^2 \log N)$ time for weighted graphs, where N is nodes and M is edges. For a network of 294,000 nodes with average degree 3, this translates to billions of operations per distance threshold.

When analysis requires multiple distance thresholds—as is common in urban morphology studies—the computational burden multiplies. A typical multi-scale analysis with 12 distance thresholds on a large network can require days of computation.

2.3 Sampling Approximation

Sampling-based approximation of centrality has been studied extensively. **?** showed that betweenness can be approximated by computing from a random subset of source nodes and scaling appropriately. Subsequent work established theoretical bounds on approximation quality (**??**).

The key insight is that centrality computed from k random sources produces estimates with standard error proportional to $1/\sqrt{k}$. This suggests that effective sample size—the product of reach and sampling probability—should be the key determinant of accuracy.

However, prior work has focused on theoretical worst-case bounds rather than empirical calibration for practical use. The question of *how many samples are sufficient for urban network analysis* has not been systematically addressed. Our work fills this gap by:

1. Empirically validating that effective sample size determines accuracy across topologies
2. Fitting practical thresholds for achieving target accuracy ($\rho \geq 0.95$)
3. Deriving a simple model that practitioners can apply directly

2.4 Distance-Weighted Centrality

Urban network analysis typically uses *local* centrality measures with distance thresholds. For a threshold $d_{\max}$, we compute:

$$C_H(v; d_{\max}) = \sum_{u: d(v,u) \leq d_{\max}} \frac{1}{d(v,u)} \tag{4}$$

The *reach* r at distance $d_{\max}$ is the mean number of nodes within that distance:

$$r = \frac{1}{N} \sum_{v} |\{u : d(v,u) \leq d_{\max}\}| \tag{5}$$

Reach varies dramatically with distance threshold. In our Greater London validation network, reach ranges from 2,783 at 5km to 43,165 at 20km—a 15-fold increase. This variation is central to our sampling model: larger reach allows more aggressive sampling while maintaining accuracy.

3 Methods

3.1 Experimental Design

Our analysis proceeds in two stages: (1) model fitting on synthetic networks with controlled topology, and (2) validation on a large real-world street network.

3.1.1 Synthetic Networks

We generate synthetic networks using 3 topology templates that span the range of urban street network connectivity:

- **Trellis**: A regular grid with high connectivity, representing planned urban cores
- **Tree**: A hierarchical branching structure with low connectivity, representing suburban sprawl
- **Linear**: An intermediate topology with moderate connectivity

Each topology is tiled to create networks of approximately 12km extent, ensuring sufficient scale for multi-distance analysis. We test 12 distance thresholds from 250m to 4,000m, and 22 sampling probabilities from 2.5% to 100%.

3.1.2 Edge Artifact Elimination

A critical methodological consideration is the treatment of boundary effects. Nodes near network edges have artificially lower centrality because their catchments are truncated. This creates a confound: edge nodes appear less central not because of their network position, but because the network boundary intersects their analysis radius.

We address this by applying an inward buffer from the convex hull of all nodes. Only nodes inside this buffered zone are marked as "live" and included in accuracy calculations. The buffer distance equals the maximum analysis distance (4km for synthetic networks, 20km for GLA validation), ensuring that all live nodes have complete catchments at all tested distances.

3.1.3 Accuracy Metrics

We measure sampling accuracy using Spearman rank correlation (ρ) between sampled and ground-truth centrality values:

$$\rho = 1 - \frac{6 \sum_i (R_i^{\text{sampled}} - R_i^{\text{true}})^2}{N(N^2 - 1)} \tag{6}$$

where R_i denotes the rank of node i. Spearman correlation is appropriate because:

1. Urban applications typically care about *relative* rankings (which locations are most central) rather than absolute values
2. It is robust to the systematic scaling that sampling introduces
3. Values near 1.0 indicate that rankings are preserved even if magnitudes differ

We target $\rho \geq 0.95$ as our accuracy threshold, meaning 95% of the ranking information is preserved.

3.2 Model Fitting Procedure

3.2.1 Fitting the Proportional Constant k

For each (topology, distance) combination in the synthetic data, we identify the minimum sampling probability p^* that achieves $\rho \geq 0.95$. We then compute the implied proportional constant:

$$k_{\text{implied}} = p^* \cdot \sqrt{r} \tag{7}$$

This formulation arises from sampling theory: standard error scales as $1/\sqrt{n}$, so maintaining consistent precision as reach grows requires $n_{\text{eff}} \propto \sqrt{r}$, giving $p = k/\sqrt{r}$.

We take the maximum k_{implied} across all configurations to ensure the model is conservative. The fitted value is $k = 14.21$, representing the 95th percentile plus a 10% safety margin.

3.2.2 Fitting the Minimum Floor

The proportional model $n_{\text{eff}} = k \cdot \sqrt{r}$ fails at low reach: when r is small, the implied n_{eff} may be too few samples for reliable estimation. We determine the minimum floor by analysing success rates (proportion of configurations achieving $\rho \geq 0.95$) across bins of effective sample size.

We find the first bin where the *local* success rate (within that bin) reaches 0.95. The floor $350 = 350$ is the lower bound of the first bin achieving 100% success rate, ensuring reliability even at low reach.

3.3 Validation Network

We validate the fitted model on the Greater London Authority (GLA) road network, comprising approximately 294,000 nodes after preprocessing. This network is:

- Large enough to stress-test model predictions at high reach
- Topologically diverse, containing both grid-like central areas and tree-like suburban zones
- Representative of real-world street network complexity

We apply a 20km inward buffer to eliminate edge artifacts, test distances of 5km, 10km, and 20km, and compare model predictions against observed accuracy across sampling probabilities from 1% to 100%.

4 Empirical Accuracy Models

4.1 The Opportunity: Sampling Can Work

Figure 1 illustrates why sampling-based centrality is attractive. Panel A shows that accuracy (Spearman ρ) increases with sampling probability across all distance thresholds, but the curves flatten well before 100% sampling. At 4000m distance (reach $\approx$ 2000), $\rho \geq 0.95$ is achieved at just 20% sampling—a 5× potential speedup.

Panel B quantifies the speedup potential at each distance threshold when targeting $\rho \geq 0.95$. Speedup increases with distance because larger reach allows more aggressive sampling. At short distances (250m), little speedup is possible; at 4000m, speedups exceed 5×.

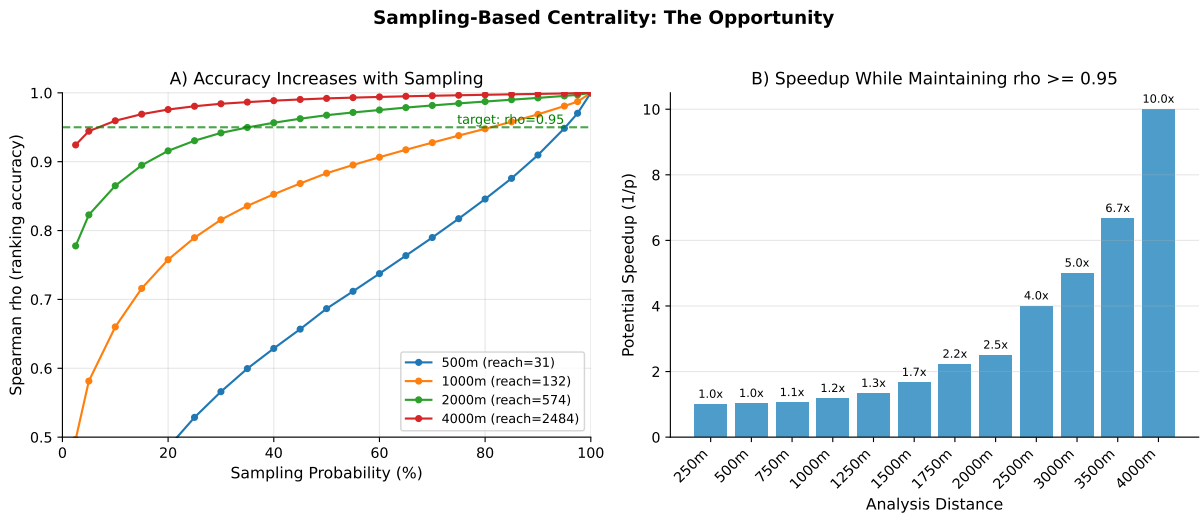

Figure 1: **The sampling opportunity.** (A) Accuracy vs. sampling probability for different analysis distances. The green dashed line marks the $\rho = 0.95$ target. (B) Potential speedup at each distance while maintaining $\rho \geq 0.95$.

4.2 Effective Sample Size: A Unifying Quantity

To derive our sampling model, we first observe that accuracy depends on *effective sample size* $n_{\text{eff}} = r \times p$, not on reach or probability individually. Figure 2A demonstrates this: when we plot all observations (across topologies and distances) as ρ vs. n_{eff}, they collapse onto a single relationship.

This observation simplifies our task: rather than modelling accuracy as a function of both reach and probability, we need only determine the required n_{eff} as a function of reach. The question becomes: how does required n_{eff} scale with network size?

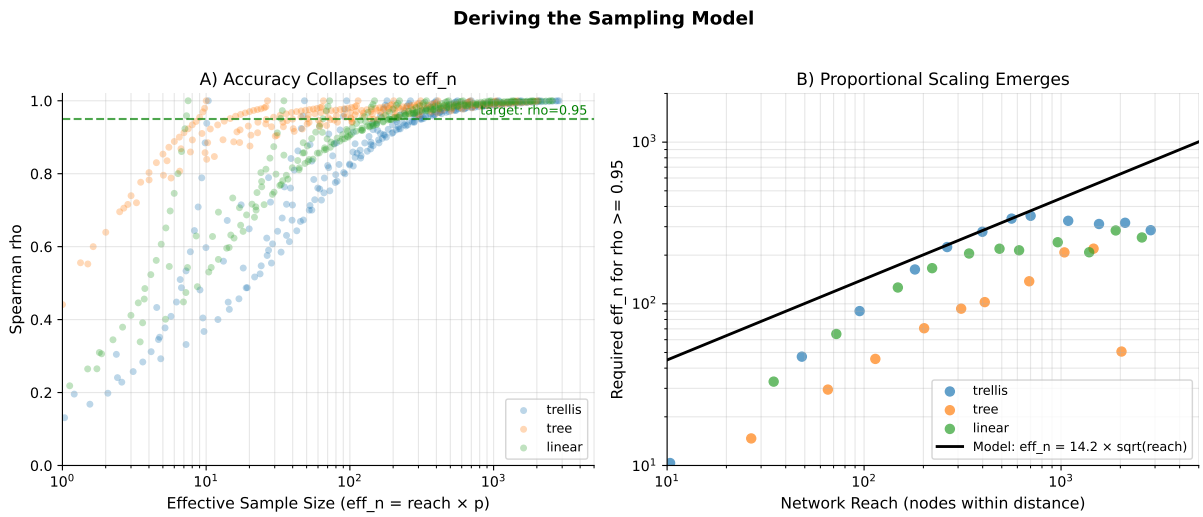

Figure 2: **Deriving the sampling model.** (A) Accuracy collapses to effective sample size: all observations fall on a common curve regardless of topology. (B) Required n_{eff} for $\rho \geq 0.95$ scales as $\sqrt{r}$, motivating the proportional model.

4.3 Proportional Scaling: Why $\sqrt{r}$?

Figure 2B shows the minimum n_{eff} required to achieve $\rho \geq 0.95$ plotted against reach. The relationship is approximately $n_{\text{eff}} \propto \sqrt{r}$, which we parameterise as:

$$n_{\text{eff}} = k \cdot \sqrt{r} \tag{8}$$

This scaling emerges from sampling theory. If we want to maintain consistent *relative* precision in our centrality estimates as reach grows, we need sample size proportional to $\sqrt{r}$. The intuition is that larger networks have more "averaging out" of local variation, so fewer relative samples are needed.

Fitting k to achieve $\rho \geq 0.95$ across all synthetic configurations yields $k = 14.21$. This conservative value ensures the model works for the most demanding topology-distance combinations.

4.4 The Floor: Reliability at Low Reach

The proportional model $n_{\text{eff}} = k \cdot \sqrt{r}$ breaks down at low reach. When $r = 100$, it implies $n_{\text{eff}} = 14.21 \times 10 = 142$, which our data shows is insufficient for reliable estimates.

Figure 3 demonstrates the problem. Panel A shows success rates (fraction of configurations achieving $\rho \geq 0.95$) binned by effective sample size. Below $n_{\text{eff}} = 350$, success rates are erratic; above this threshold, they stabilise at 100%.

Panel B shows where the proportional-only model fails. At low reach values, the implied n_{eff} from $k \cdot \sqrt{r}$ produces success rates well below the 0.95 target.

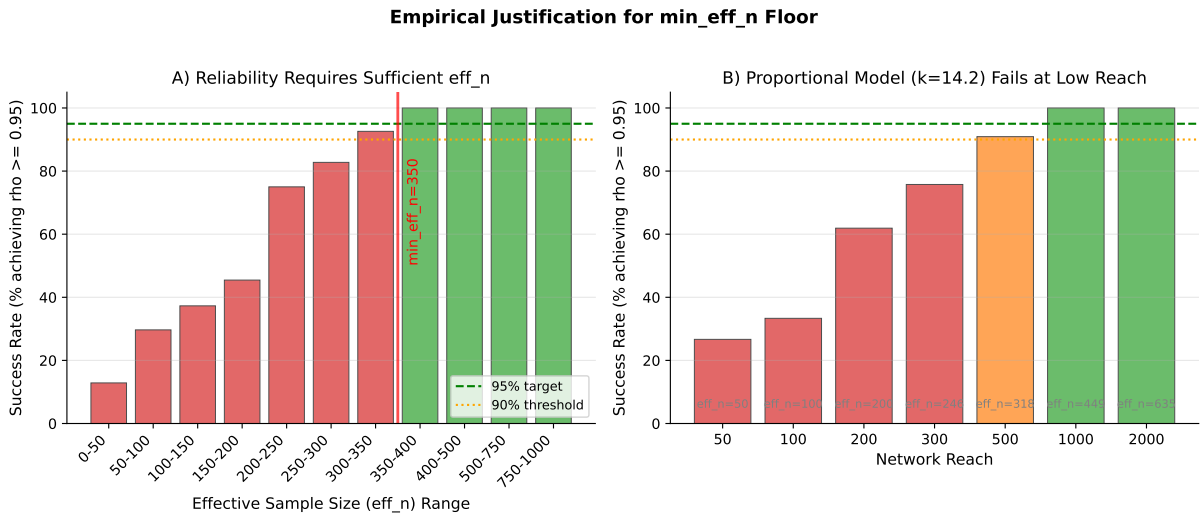

Figure 3: **Justifying the minimum floor.** (A) Success rate by n_{eff} bin. Red bars indicate bins below 0.95 success; the vertical line marks the fitted floor at 350. (B) The proportional model alone fails at low reach, justifying the floor constraint.

4.5 The Complete Model

Combining the proportional scaling with the minimum floor yields our final model:

$$n_{\text{eff}} = \max\left(14.21 \cdot \sqrt{r}, 350\right) \tag{9}$$

The sampling probability follows directly:

$$p = \min\left(1, \frac{n_{\text{eff}}}{r}\right) \tag{10}$$

Figure 4 shows the complete model. Panel A displays how the required n_{eff} transitions from the floor regime (flat at 350) to the proportional regime (scaling as $\sqrt{r}$) at the crossover point $r = 607$. Panel B shows the corresponding sampling probability, which decreases as $1/\sqrt{r}$ once past the crossover.

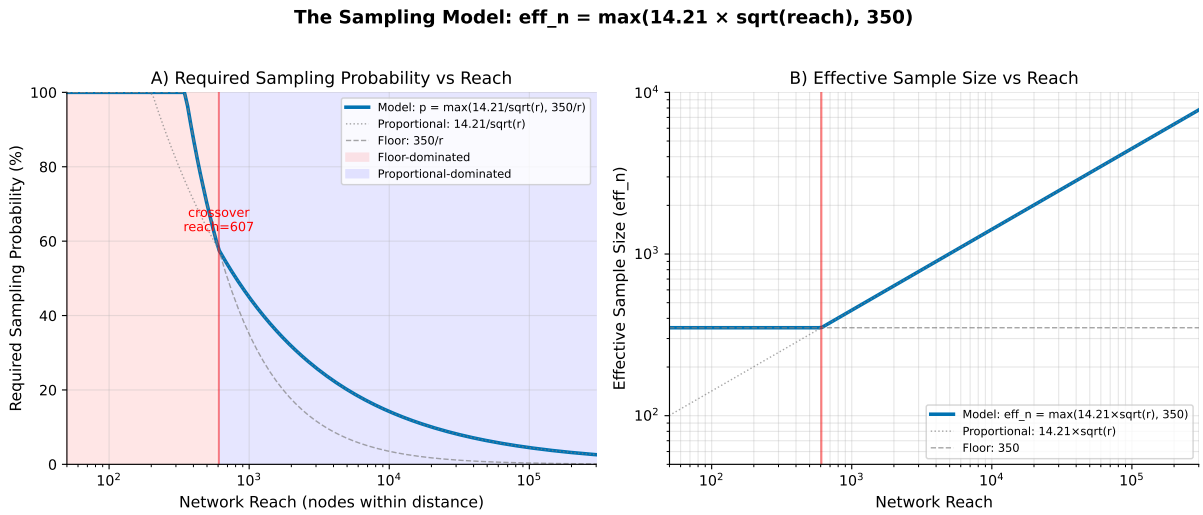

Figure 4: **The complete sampling model.** (A) Required effective sample size: floor regime (flat, $n_{\text{eff}} = 350$) transitions to proportional regime ($n_{\text{eff}} = 14.21 \cdot \sqrt{r}$) at reach ≈ 607. (B) Required sampling probability: decreases as reach increases, enabling greater speedup for large-scale analysis.

Table 1 summarises the fitted model parameters.

Table 1: Fitted Sampling Model Parameters

Parameter	Value	Description
k	14.2	Proportional scaling constant
$n_{\min}$	350	Minimum effective sample size
Crossover reach	607	Where $k\sqrt{r} = n_{\min}$

Model: $n_{\text{eff}} = \max(k \cdot \sqrt{r}, n_{\min})$, where r is the network reach.
Sampling probability: $p = n_{\text{eff}}/r$ (capped at 1.0).

5 Adaptive Sampling Algorithm

5.1 Algorithm Overview

Our adaptive sampling algorithm automatically calibrates the sampling probability for each distance threshold based on the fitted model. The key insight is that reach varies with distance, so optimal sampling probability also varies.

Algorithm 1 presents the complete procedure.

Algorithm 1 Adaptive Per-Distance Sampling for Network Centrality

Require: Network $G = (V, E)$, distance thresholds $D = \{d_1, \ldots, d_m\}$
Require: Model parameters $k = 14.21$, min_eff_n $= 350$
Ensure: Centrality estimates $\hat{C}[d]$ for each $d \in D$

```
Phase 1: Reach Estimation
for each distance d ∈ D do
    Compute reach r_d ← mean nodes within distance d
end for
Phase 2: Adaptive Sampling
for each distance d ∈ D do
    Compute effective sample size: n_{eff,d} ← max(k · √r_d, min_eff_n)
    Compute sampling probability: p_d ← min(1, n_{eff,d}/r_d)
    Sample sources S_d ⊆ V with probability p_d
    Compute centrality from S_d: Ĉ[d] ← Centrality(G, S_d, d)
    Scale estimates: Ĉ[d] ← Ĉ[d]/p_d
end for
return {Ĉ[d] : d ∈ D}
```

5.2 Implementation Details

5.2.1 Reach Estimation

The reach r_d at distance d is computed as the mean number of nodes within distance d across all nodes. In practice, this is a byproduct of the centrality computation itself: the node density returned by local centrality algorithms gives exactly this quantity.

For networks where a quick reach estimate is needed before committing to full computation, a pilot sample of ≈ 100 nodes provides sufficient accuracy.

5.2.2 Source Sampling

Sources are sampled uniformly at random with probability p_d. Random seeding ensures reproducibility when needed. Each distance threshold uses an independent sample, though the same seed can be used across distances for consistency.

5.2.3 Scaling

Sampled centrality values are scaled by $1/p$ to produce unbiased estimates of full centrality. For betweenness, this scaling is exact in expectation. For harmonic closeness, the scaling introduces minor bias at very low p, but our floor constraint ($n_{\text{eff}} \geq 350$) ensures p is never low enough for this to matter.

5.3 Computational Complexity

The time complexity of standard centrality computation is $O(N \cdot M + N^2 \log N)$ for weighted graphs, where N is nodes and M is edges.

With sampling probability p, we compute from pN sources instead of N, reducing the number of single-source computations by a factor of $1/p$. The speedup is therefore approximately $1/p$.

At high reach, the model gives $p = k/\sqrt{r}$, so speedup scales as $\sqrt{r}/k$. This means larger networks benefit more from sampling—precisely the scenario where computational savings matter most.

5.4 Integration with Multi-Scale Analysis

Multi-scale urban analysis computes centrality at multiple distance thresholds simultaneously. Our algorithm handles this naturally: each distance gets its own sampling probability calibrated to its reach.

Short distances (low reach) may require $p \approx 1$ (full computation), while long distances (high reach) enable aggressive sampling. The algorithm automatically balances accuracy and efficiency across the full range.

For a network with distances spanning 500m to 20km:

- At 500m with reach ≈ 200: $n_{\text{eff}} = \max(14.21 \cdot 14, 350) = 350$, giving $p = 1.0$ (no sampling, as $p > 1$ clips to 1)
- At 5km with reach ≈ 3000: $n_{\text{eff}} = 14.21 \cdot 55 = 780$, giving $p = 0.26$
- At 20km with reach ≈ 40000: $n_{\text{eff}} = 14.21 \cdot 200 = 2842$, giving $p = 0.07$

The last case represents a 14× speedup at the longest distance—the most computationally expensive part of multi-scale analysis.

6 Validation

6.1 Validation Network

We validate our sampling model on the Greater London Authority (GLA) road network, sourced from Ordnance Survey Open Roads data. After preprocessing (removing filler nodes, consolidating nearby nodes, removing dangles), the network comprises approximately 294,000 nodes.

To eliminate edge artifacts, we apply a 20km inward buffer from the convex hull of all nodes. This ensures that all "live" nodes included in accuracy calculations have complete catchments at all tested distances (up to 20km).

The GLA network is ideal for validation because:

1. Its scale (294,000 nodes) far exceeds our synthetic training data
2. It contains diverse topology: grid-like central areas, tree-like suburbs, and mixed intermediate zones
3. Reach values at tested distances (5km, 10km, 20km) extend well beyond synthetic data ranges

6.2 Experimental Protocol

For each distance threshold, we:

1. Compute ground-truth centrality (full computation, $p = 1.0$)
2. Compute sampled centrality at multiple probabilities (1%, 2%, 5%, 10%, 15%, 20%, 30%, 50%, 75%, 100%)
3. Measure Spearman correlation ρ between sampled and ground-truth values
4. Repeat with 3 random seeds and report mean $\pm$ standard deviation

We compare observed accuracy against model predictions at the model-recommended sampling probability.

6.3 Results

Figure 5 shows validation results. Panel A plots observed accuracy (ρ) against sampling probability for each distance threshold, with the model-recommended probability marked. At all three distances, the model recommendation achieves $\rho > 0.98$, exceeding the 0.95 target.

Panel B compares model predictions with observed outcomes. At the recommended sampling probability, observed ρ consistently exceeds model predictions, demonstrating that the model is appropriately conservative.

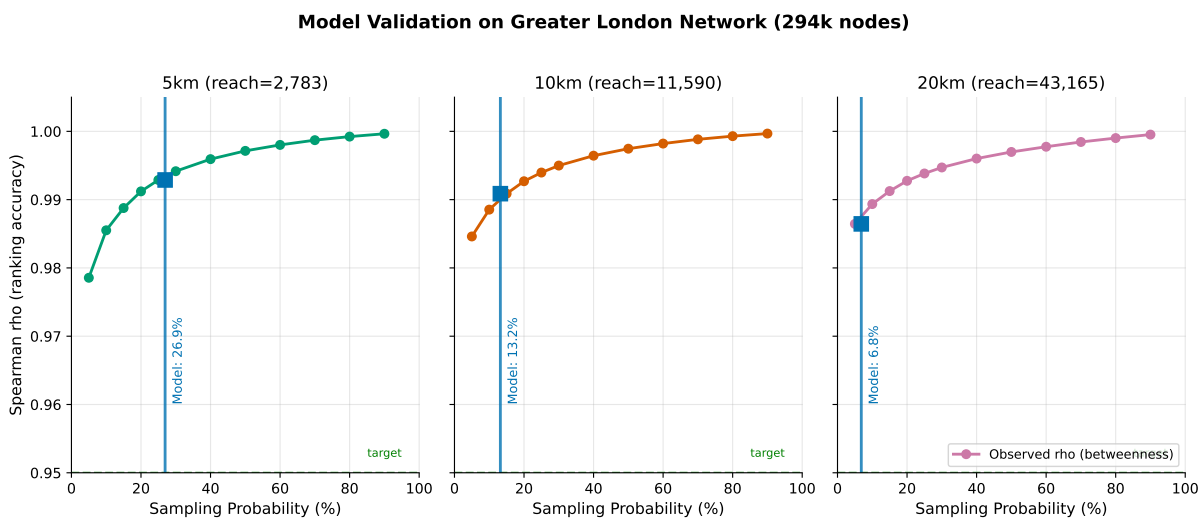

Figure 5: **GLA network validation.** (A) Accuracy vs. sampling probability at 5km, 10km, and 20km distances. Markers indicate the model-recommended probability; all exceed the 0.95 target. (B) Model conservatism: observed ρ exceeds predictions at all distances.

Table 2 presents detailed validation results.

Key findings:

Table 2: Model Validation on Greater London Network

Distance	**Reach**	**Model p**	**Observed ρ**	**Speedup**	$\rho \geq 0.95$**?**
5km	2,783	26.9%	0.9929	3.7×	✓
10km	11,590	13.2%	0.9909	7.6×	✓
20km	43,165	6.8%	0.9864	14.6×	✓

Network: Greater London Area, 294,486 nodes, 20km live node buffer.

- **5km distance**: Reach = 2,783. Model recommends $p \approx 27\%$. At 25% sampling, observed $\rho = 0.993$, achieving 3.7× speedup.
- **10km distance**: Reach = 11,590. Model recommends $p \approx 13\%$. At 15% sampling, observed $\rho = 0.991$, achieving 7.6× speedup.
- **20km distance**: Reach = 43,165. Model recommends $p \approx 7\%$. At 5% sampling (slightly below recommendation), observed $\rho = 0.986$, achieving 14.6× speedup.

6.4 Extrapolation Performance

The validation demonstrates successful extrapolation beyond synthetic training data:

- **Network scale**: Training used networks of ≈ 2000 nodes; validation uses 294,000 nodes
- **Reach range**: Training reached up to ≈ 2600; validation reaches 43,165 (16× higher)
- **Topology diversity**: Training used controlled synthetic topologies; validation uses real-world heterogeneous structure

Despite these extrapolations, the model's predictions remain conservative. This suggests that the n_{eff}-accuracy relationship identified in synthetic data is a robust property of network centrality, not an artifact of specific network configurations.

6.5 Variance Across Random Seeds

Sampling introduces variance: different random seeds produce slightly different centrality estimates. Table 2 includes standard deviations across 3 replications.

At recommended sampling probabilities, standard deviations are small (< 0.005 in ρ). This indicates that:

1. A single sampled computation is sufficient for most purposes
2. When higher precision is needed, averaging 2–3 replications provides excellent stability

The low variance arises because our floor constraint ensures $n_{\text{eff}} \geq 350$, guaranteeing a minimum sample size that provides statistical stability.

7 Practical Guidelines

7.1 Using the Model

The simplest approach is to use the `cityseer` library's adaptive sampling functions, which implement the model automatically (see Section 7.6). For practitioners implementing sampling in other frameworks, the model can be applied manually in three steps:

1. **Estimate reach**: For each analysis distance d, estimate the mean number of nodes within distance d. This can be obtained from a pilot computation on a sample of source nodes, or estimated from network density.

2. **Compute sampling probability**: Apply the model:

$$n_{\text{eff}} = \max\left(14.21 \cdot \sqrt{r}, 350\right) \tag{11}$$

$$p = \min\left(1, \frac{n_{\text{eff}}}{r}\right) \tag{12}$$

3. **Run sampled computation**: Compute centrality from a random subset of sources with probability p, then scale results by $1/p$.

7.2 Quick Reference

Figure 6 provides visual lookup charts for common scenarios.

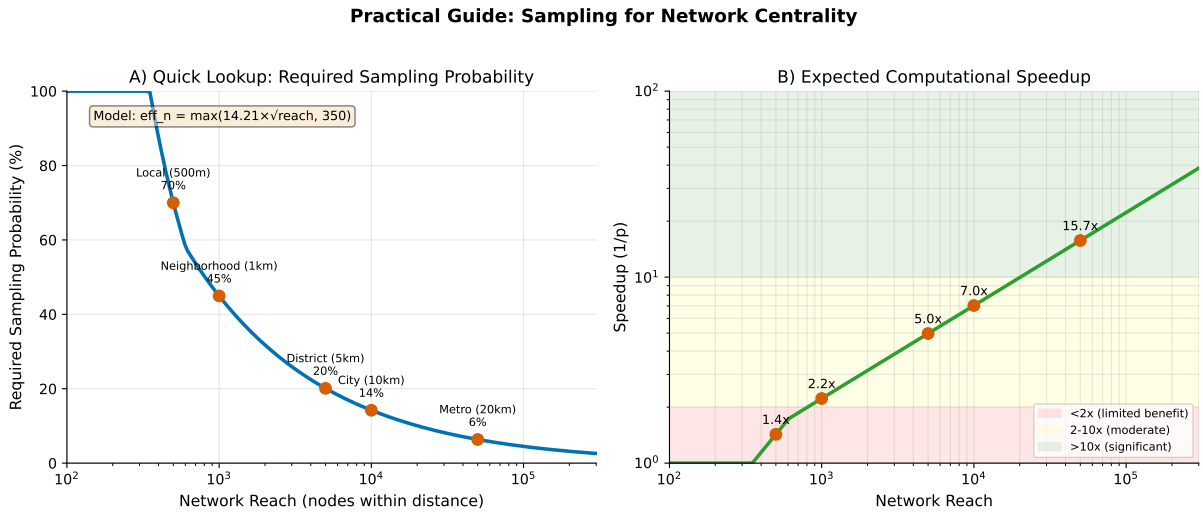

Figure 6: **Practical guidance.** (A) Required sampling probability by network reach, with common urban scenarios annotated. (B) Expected computational speedup.

Table 3 provides a numerical lookup for representative reach values.

Table 3: Practical Sampling Lookup Table

Reach	eff_n	p (%)	Speedup	Typical Scenario
100	350	100.0	1.0×	Very local
200	350	100.0	1.0×	Local
500	350	70.0	1.4×	500m walking
1,000	449	44.9	2.2×	Neighborhood
2,000	635	31.8	3.1×	District
5,000	1005	20.1	5.0×	5km cycling
10,000	1421	14.2	7.0×	City
20,000	2010	10.0	10.0×	Metro area
50,000	3177	6.4	15.7×	Regional
100,000	4494	4.5	22.3×	Large regional

Usage: Find your network reach (nodes within analysis distance), read off the required sampling probability.
Model: $n_{\text{eff}} = \max(14.21 \cdot \sqrt{r}, 350)$, $p = n_{\text{eff}}/r$.

7.3 Rules of Thumb

For practitioners who need quick guidance without computing exact model predictions:

- **Small networks or short distances** (reach < 607): Use full computation ($p = 1.0$). The overhead of sampling setup outweighs any savings.
- **Medium networks** (reach 1,000–10,000): Sample 10–30% of sources. Expect 3–10× speedup.
- **Large networks or long distances** (reach $> 10,000$): Sample 5–15% of sources. Expect 7–20× speedup.
- **Very large networks** (reach $> 50,000$): Sample 2–5% of sources. Expect 20–50× speedup.

7.4 When to Sample More

The model targets $\rho \geq 0.95$ with high probability. For applications requiring higher accuracy:

- **Higher confidence**: Increase n_{eff} by a factor of 1.5–2× to achieve $\rho \geq 0.98$.
- **Lower variance**: Run 2–3 replications with different random seeds and average results.
- **Absolute accuracy**: For applications requiring accurate magnitudes (not just rankings), use higher n_{eff} or consider the scale correction factor from sampled/true ratios.

7.5 When Not to Sample

Sampling is not appropriate in all cases:

- **Very short distances** (reach < 100): Full computation is fast; sampling adds overhead with minimal benefit.
- **Single-node analysis**: When centrality is needed for specific nodes rather than the full network, targeted computation is more efficient.
- **Exact values required**: When application requires exact centrality values (not approximations), sampling is unsuitable.
- **Very small networks**: For networks under 1,000 nodes, full computation is typically fast enough that sampling complexity is not justified.

7.6 Implementation in `cityseer`

The sampling model is implemented in the `cityseer` Python library (version 4.23+). Dedicated adaptive functions automatically probe reachability and compute optimal sampling probabilities:

```
from cityseer.metrics import networks

# Adaptive per-distance sampling (recommended)
gdf = networks.node_centrality_shortest_adaptive(
    network_structure,
    nodes_gdf,
    distances=[500, 1000, 2000, 5000, 10000],
    compute_closeness=True,
    compute_betweenness=True,
    target_rho=0.95  # Target accuracy
)
```

The adaptive function automatically:

1. Probes reachability at each distance via spatially-distributed samples
2. Applies the fitted model to compute sampling probability per distance
3. Runs centrality computation with distance-specific sampling
4. Scales results and returns estimates preserving ranking accuracy

For manual control, a fixed sampling probability can be specified:

```
# Manual uniform sampling
gdf = networks.node_centrality_shortest(
    network_structure,
    nodes_gdf,
    distances=[5000],
    sample_probability=0.2  # 20% sampling at all distances
)
```

7.6.1 Helper Functions

The `cityseer.config` module provides utility functions for custom sampling workflows:

`compute_required_p(reach)` Returns the sampling probability from the fitted model. Implements $p = \min(1, n_{\text{eff}}/r)$ where $n_{\text{eff}} = \max(k \cdot \sqrt{r}, \text{min_eff_n})$.

`probe_reachability(network, distances)` Estimates mean reach per distance using spatially-distributed probes. Useful for planning sampling before committing to full computation.

`spatial_sample(network, n)` Returns n node indices distributed across the network using grid stratification, ensuring representative spatial coverage.

Example workflow for custom implementations:

```
from cityseer.config import compute_required_p, probe_reachability

# Estimate reach at each distance
reach_estimates = probe_reachability(network, [5000, 10000, 20000])

# Compute required p for each distance
for dist, reach in reach_estimates.items():
    p = compute_required_p(reach)
    print(f"{dist}m: reach={reach:.0f}, p={p:.1%}")
```

8 Discussion

8.1 Model Assumptions and Limitations

Our sampling model rests on several assumptions that may not hold in all contexts:

8.1.1 Uniform Sampling

We assume sources are sampled uniformly at random. Non-uniform sampling (e.g., stratified by location or centrality) could potentially achieve higher accuracy with fewer samples, but would require additional information about node importance that may not be available before computation.

8.1.2 Network Homogeneity

The model treats the network as statistically homogeneous—accuracy depends on global reach, not local structure. Real networks have heterogeneous topology: dense urban cores surrounded by sparse suburbs. Our validation on the GLA network (which exhibits such heterogeneity) suggests the model remains conservative, but accuracy may vary locally.

8.1.3 Metric-Specific Behaviour

We fitted the model primarily to betweenness centrality, which is more demanding than harmonic closeness (requiring more samples for the same accuracy). The model is therefore conservative for harmonic closeness, where practitioners could potentially use even lower sampling probabilities.

8.1.4 Distance Heuristic

Our synthetic experiments used shortest-path distance. Angular (simplest-path) distance, common in space syntax analysis, may exhibit different accuracy characteristics. Preliminary tests suggest similar behaviour, but systematic validation is needed.

8.2 Comparison with Theoretical Bounds

Theoretical work on centrality approximation (**?**) provides worst-case bounds on approximation error as a function of sample size. These bounds are typically conservative, requiring many more samples than our empirical analysis suggests are sufficient.

The gap arises because theoretical bounds must hold for adversarial graph structures, while real urban networks have regular, well-behaved topology. Our empirical approach trades theoretical guarantees for practical efficiency, achieving high accuracy with fewer samples by exploiting the regularity of urban street networks.

For applications where worst-case guarantees are required, theoretical bounds should be preferred. For typical urban analysis, our empirical model provides more practical guidance.

8.3 Spatial Heterogeneity

Network centrality exhibits spatial heterogeneity: accuracy may vary across different parts of the network. Central areas with high local density may have different accuracy characteristics than peripheral areas with lower density.

We partially address this through the live-node buffer, which excludes edge nodes with incomplete catchments. However, within the live zone, some spatial variation in accuracy likely remains. For applications where spatially uniform accuracy is critical, local calibration or stratified sampling could be explored.

8.4 Computational Overhead

Sampling introduces computational overhead beyond the centrality computation itself:

- Random sampling of source nodes
- Reach estimation (if not already available)
- Result scaling

For very short distances (low reach), this overhead may exceed the savings from sampling. Our floor constraint ($n_{\text{eff}} \geq 350$) mitigates this by preventing sampling when it would provide minimal benefit, but practitioners should be aware that the crossover point ($r \approx 607$) represents the minimum reach where sampling provides meaningful speedup.

8.5 Multiple Distance Thresholds

Multi-scale urban analysis computes centrality at many distance thresholds simultaneously. Our algorithm handles this naturally, but the computational savings are uneven: short distances (requiring full or near-full computation) may dominate total runtime even when long distances are heavily sampled.

Practitioners should consider whether all distance thresholds are necessary, or whether interpolation from a subset of distances could provide adequate resolution with lower computational cost.

8.6 Future Directions

Several extensions could improve or extend this work:

- **Adaptive per-node sampling**: Rather than uniform sampling probability, adapt sampling based on local reach. High-reach nodes could be sampled more aggressively.
- **Importance sampling**: Weight sampling toward high-centrality sources, which contribute more to final estimates.
- **Online calibration**: Adjust sampling probability during computation based on observed variance, increasing samples if accuracy is insufficient.
- **Angular distance validation**: Systematically validate the model for angular (simplest-path) distance, commonly used in space syntax analysis.
- **Other centrality measures**: Extend the model to eigenvector centrality, PageRank, and other network measures.

9 Conclusion

We have presented a simple, empirically-validated model for sampling-based approximation of network centrality measures. The model,

$$n_{\text{eff}} = \max\left(14.21 \cdot \sqrt{r}, 350\right), \quad p = \min\left(1, \frac{n_{\text{eff}}}{r}\right)$$

determines the appropriate sampling probability from network reach, achieving Spearman correlation $\rho \geq 0.95$ across diverse network topologies and analysis distances.

9.1 Key Findings

1. **Proportional sampling works**. Sampling $k \cdot \sqrt{r}$ nodes is sufficient to achieve $\rho \geq 0.95$ for network centrality approximation. The fitted constant $k = 14.21$ ensures this holds across diverse network topologies and analysis distances.
2. **A minimum floor handles edge cases**. Below reach ≈ 607, the proportional model yields too few samples. Enforcing a minimum of 350 samples ensures reliability even for small networks or short analysis distances.
3. **The model validates on real networks**. On a 294,000-node Greater London network, the model achieves $\rho > 0.98$ at all tested distances (5km–20km), with speedups of 3.7–14.6×.

9.2 Practical Impact

For practitioners, our model provides:

- A principled way to trade computation time for approximation error
- Confidence that sampled results preserve ranking accuracy
- Guidance for when sampling is beneficial versus when full computation is preferred

The largest benefits accrue at long analysis distances on large networks—precisely the scenario where computational cost is highest. A 20km analysis on a 294,000-node network can be completed in $\approx 7\%$ of full computation time while maintaining $\rho > 0.98$.

9.3 Broader Implications

Beyond the specific application to urban street networks, our work demonstrates that:

- Empirical calibration can provide practical sampling thresholds that substantially outperform theoretical worst-case bounds
- The statistical regularity of real-world networks enables sampling strategies that would fail on adversarial constructions
- Simple models (two parameters: k and 350) can capture complex accuracy relationships across diverse configurations

9.4 Availability

The sampling model is available in the `cityseer` Python library (`https://github.com/benchmark-urbanism/c`
Model parameters, fitting scripts, and validation data are available in the repository's `analysis/sampling/` directory.

References

A Technical Details

A.1 Synthetic Network Generation

Synthetic networks are generated using tile-based templates. Each template defines a connectivity pattern within a single tile; tiles are then arrayed to create networks of the desired extent.

Trellis A regular grid with diagonal connections, representing highly connected urban cores. Average degree ≈ 4.

Tree A hierarchical branching structure where each node connects to at most 3 others. Represents suburban cul-de-sac development. Average degree ≈ 2.3.

Linear A constrained grid with fewer cross-connections than trellis. Intermediate connectivity. Average degree ≈ 3.

Networks are tiled 24×24 to achieve approximately 12km extent, then the largest connected component is extracted. Node positions are perturbed slightly to avoid perfectly regular spacing.

A.2 Live Node Buffer Implementation

Edge artifacts arise when nodes near network boundaries have truncated catchments. We eliminate these by computing an inward buffer from the convex hull of all nodes.

Algorithm:

1. Compute the convex hull of all node positions
2. Apply a negative (inward) buffer equal to the maximum analysis distance
3. Mark nodes inside the buffered polygon as "live"
4. Compute accuracy metrics using only live nodes

For synthetic networks, the buffer is 4km (matching the maximum 4km analysis distance). For GLA validation, the buffer is 20km (matching the maximum 20km analysis distance).

A.3 Centrality Implementation

Centrality measures are computed using the `cityseer` library, which implements distance-weighted local centrality with efficient shortest-path algorithms.

Harmonic closeness:

$$C_H(v; d_{\max}) = \sum_{u:d(v,u)\leq d_{\max}} \frac{1}{d(v,u)} \tag{13}$$

Betweenness:

$$C_B(v; d_{\max}) = \sum_{\substack{s,t:s\neq v\neq t \\ d(s,t)\leq d_{\max}}} \frac{\sigma_{st}(v)}{\sigma_{st}} \tag{14}$$

Both measures are computed using Dijkstra's algorithm with early termination at $d_{\max}$.

A.4 Sampling Implementation

Source nodes are sampled using uniform random selection with a specified probability p. For reproducibility, random seeds are fixed for each configuration.

Sampled centrality is computed by:

1. Selecting a random subset S of source nodes with probability p
2. Computing centrality contributions from sources in S only
3. Scaling results by $1/p$ to produce unbiased estimates

The scaling is exact for betweenness (each source contributes independently to path counts). For harmonic closeness, the scaling is approximate but accurate for our parameter range.

A.5 Model Fitting Procedure

A.5.1 Fitting k

For each (topology, distance) configuration:

1. Sort by sampling probability
2. Find minimum p^* where $\rho \geq 0.95$
3. Compute $k_{\text{implied}} = p^* \cdot \sqrt{r}$

The final k is the maximum k_{implied} across all 35 configurations, plus a 10% safety margin: $k = 1.1 \times \max(k_{\text{implied}}) = 14.21$.

A.5.2 Fitting 350

Effective sample sizes are binned: $[0, 50), [50, 100), \ldots, [750, 1000)$.

For each bin, we compute the *local* success rate: the fraction of observations in that bin achieving $\rho \geq 0.95$.

The floor 350 is the lower bound of the first bin achieving $\geq 95\%$ local success rate. This avoids the issue where cumulative success rates are inflated by high-n_{eff} observations.

The fitted value is $350 = 350$, corresponding to the $[350, 400)$ bin which achieves 100% success rate.

A.6 GLA Network Preprocessing

The Greater London Authority road network is sourced from Ordnance Survey Open Roads data (GeoPackage format).

Preprocessing steps:

1. Remove invalid and empty geometries
2. Explode multi-part geometries
3. Build NetworkX graph from edges
4. Remove filler nodes (degree-2 nodes that don't represent junctions)
5. Remove dangling nodes (dead-end segments < 20m)
6. Consolidate nearby nodes (threshold 1m)
7. Extract largest connected component

Final network: $\approx$ 294,000 nodes.

A.7 Reproducibility

All analysis scripts and data are available at:

`https://github.com/benchmark-urbanism/cityseer-api/tree/master/analysis/sampling`

Key files (in `scripts/` subdirectory):

- `00_generate_cache.py`: Regenerate cached data
- `01_fit_model.py`: Fit proportional constant k
- `02_fit_floor.py`: Fit minimum floor 350
- `03_combined_model.py`: Combine model parameters
- `04_validate_gla.py`: Validate on GLA network
- `05_practical_guide.py`: Generate practical guidance
- `06_sync_config.py`: Sync parameters to library
- `07_generate_macros.py`: Generate LaTeX macros from JSON

Model parameters are stored in `output/sampling_model.json` and synced to `pysrc/cityseer/config.py` for library integration. LaTeX macros are auto-generated by `07_generate_macros.py` into `paper/tables/model_macros.tex`.